

Lec 4

Basic of Internet

Date 19/1/24
Page

IP address (in detail)

IPv4 32-bit hi kyu hota h? becoz a standard is set, so that 2^{32} it does not get confused & read only 32 bits.

→ IPv6: 128 bits

extended) L in hexadecimal

$\times 64 \rightarrow 128$ ✓ 2^{128} (planned for 200 yrs future)

becoz of no. of devices are more !!

min) 0000.0000.0000.0000.0000.0000.0000.0000
max) ffff.ffff.ffff.ffff.ffff.ffff.ffff.ffff
4 bits 16 bits Total 8

15 is the last decimal value in hexadecimal format to store that we need 16 bits

$$16 \times 8 = 128 \text{ bits.}$$

This format is for human, computer reads in binary format.

IPv5 was not implemented becoz it was not sufficient for future.

How class of IP address were implemented

→ Class A in binary representation

8 bits. 8 bits. 8 bits. 8 bits

first bit is set 0 _____

0 0 0 0 0 0 0 0

0 1 1 1 1 1 1 1

max $2^7 = 128$

that's why 0-127 values

→ Class B

first bit set to 1

second bit set to 0

1 0

8 bits 8 bits 8 bits

$2^6 = 64$

6 left

$128 - 191$

$10\ 000000 = 128$

$10\ 111111 = 191$

→ Class C

first set to 1, second to 1, third to 0

110 5 left

$2^5 = 32$

$192 - 223$

$110\ 00000 = 192$

$110\ 11111 = 223$

→ Class D

first set to 1, second to 1, third to 1, 4th to 0

1 1 1 0 4 left

$$2^4 = 16$$

$$\hookrightarrow 224 - 239$$

$$11100000 = 224$$

$$11101111 = 239$$

→ Class E

$$11110000 = 240 \quad \boxed{2^4 = 16}$$

$$11111111 = 255$$

$$\hookrightarrow 240 - 255$$

fixed 4 left

Summary

Class A : 0

Class B : 10

Class C : 110

Class D : 1110

Class E : 1111

IANA → International Assigned Number Authority (distributes IP Address)



North America,

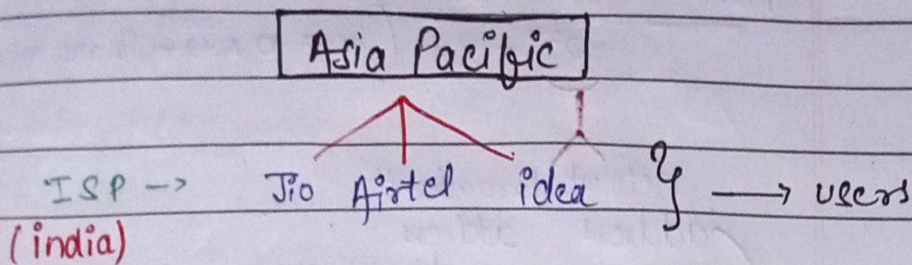
Africa, Europe,

Latin America,

Asia-Pacific.

← To 5 continents

→ country wise ISP



→ Why divided into classes ?

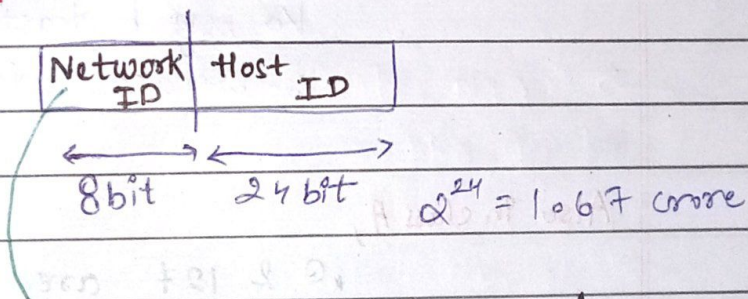
ISP ke need ke hisaab se usko IP address dena tha.

↳ crore IP address lenne wale ISP kam hai

↳ lakh IP address lenne wale zyada

↳ thousand IP lenne wale use b zyada hai.

Class A



0 -----

$2^7 = 128 \text{ network}$

(network IDs distributed to ISP)

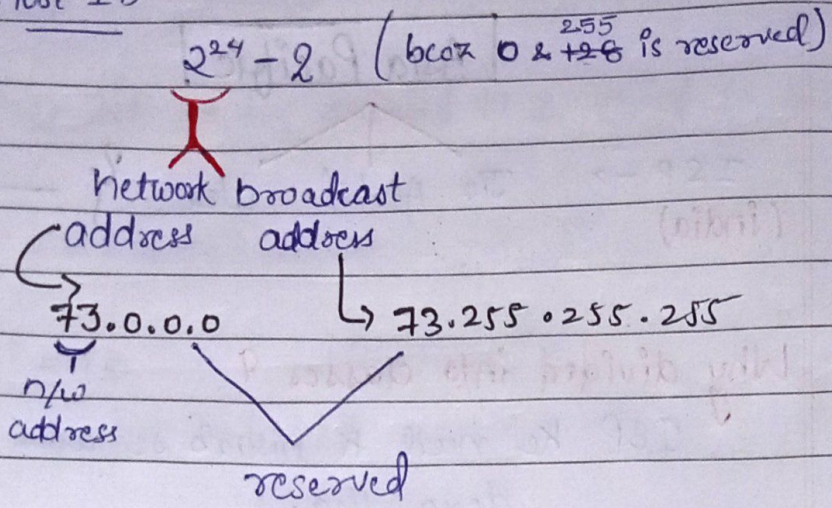
{ 73.0.0.0

{ 73.255.255.255

network ID

(0-127) - 128 n/w (har ek n/w me 1.067 host id h)

Host ID



b/w these are for public.

* Network address $\rightarrow$ jab user first time IP address request karta hai tb uske paas jo IP hota hai wo.
0.0.0.0 (reserved)

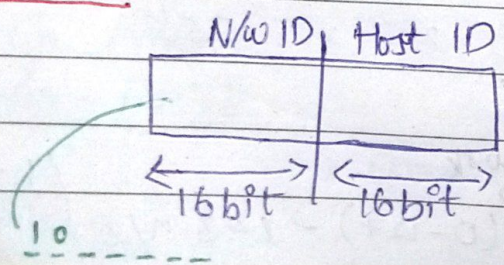
broadcast address $\rightarrow$ jab n/w ke saare devices ko msg broadcast krna ho tb broadcast address dalte h.

Also in class A,

* 0 & 127 are reserved
0.0.0.0 $\leftarrow$ loopback address

1-126 (for public) $2^7 - 2 = 126$

Class B



N/ID $2^{14} = 16000$

(1 0 are fixed in start)

H/ID

$2^{16} = 65000$

ISP needed 1 lakh (example)

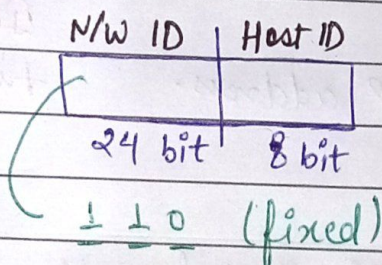
↳ allocate 2 network

① 1 - 65K IP add.

② 2 - 65K IP add.

Total no. of network in class B = $2^{16} - 2$

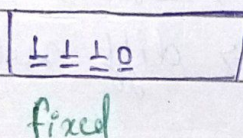
Class C



N/w ID 2^{21} (3 bits are fixed)

Host ID $2^8 - 2 = 254$

Class D

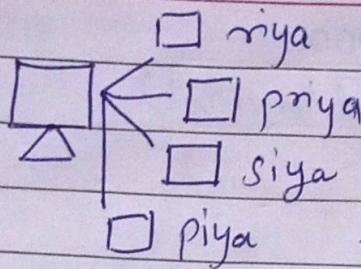


Multicasting



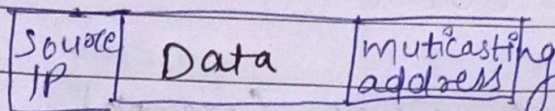
(zoom call)

$2^{28} \rightarrow$ IP address



else
system crash

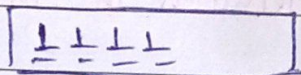
can't send to every user personally.



Single packet

in place of destination IP address

Class E



fixed

$2^{28} \rightarrow$ IP address.

reserved by IANA for future use.

$\rightarrow$ IANA distributed all 4 classes of IP address to 5 countries.

$\rightarrow$ In all classes there are also some private ranges.

Broadcasting $\rightarrow$ same network

Multicasting $\rightarrow$ diff network also